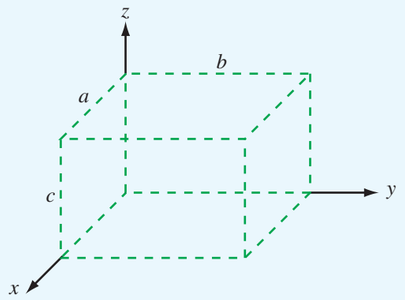


Figure P7.39 Imaginary rectangular box of Problems 7.39 and 7.40.

7.41 Given a wave with $\mathbf{E} = \hat{\mathbf{x}} E_0 \cos(\omega t - kz)$:

*(a) Calculate the time-average electric energy density

$$(w_e)_{av} = \frac{1}{T}\int_0^T w_e\, dt = \frac{1}{2T}\int_0^T \varepsilon E^2\, dt.$$

(b) Calculate the time-average magnetic energy density

$$(w_m)_{av} = \frac{1}{T}\int_0^T w_m\, dt = \frac{1}{2T}\int_0^T \mu H^2\, dt.$$

(c) Show that $(w_e)_{av} = (w_m)_{av}$.

7.42 A team of scientists is designing a radar as a probe for measuring the depth of the ice layer over the antarctic land mass. In order to measure a detectable echo due to the reflection by the ice-rock boundary, the thickness of the ice sheet should not exceed three skin depths. If $\varepsilon_r' = 3$ and $\varepsilon_r'' = 10^{-2}$ for ice and if the maximum anticipated ice thickness in the area under exploration is 1.2 km, what frequency range is useable with the radar?

336 CHAPTER 7 PLANE-WAVE PROPAGATION

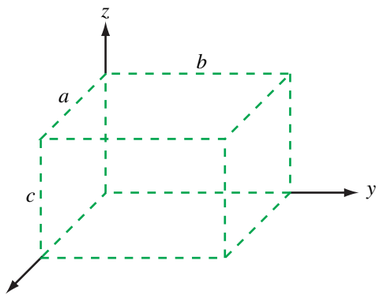


图 P7.39 问题 7.39 和 7.40 的假想矩形框。

7.41 给定一个具有 $E = \hat{x}\, E_0 \cos(\omega t - kz)$ 的波：

*(a) 计算时均电能密度

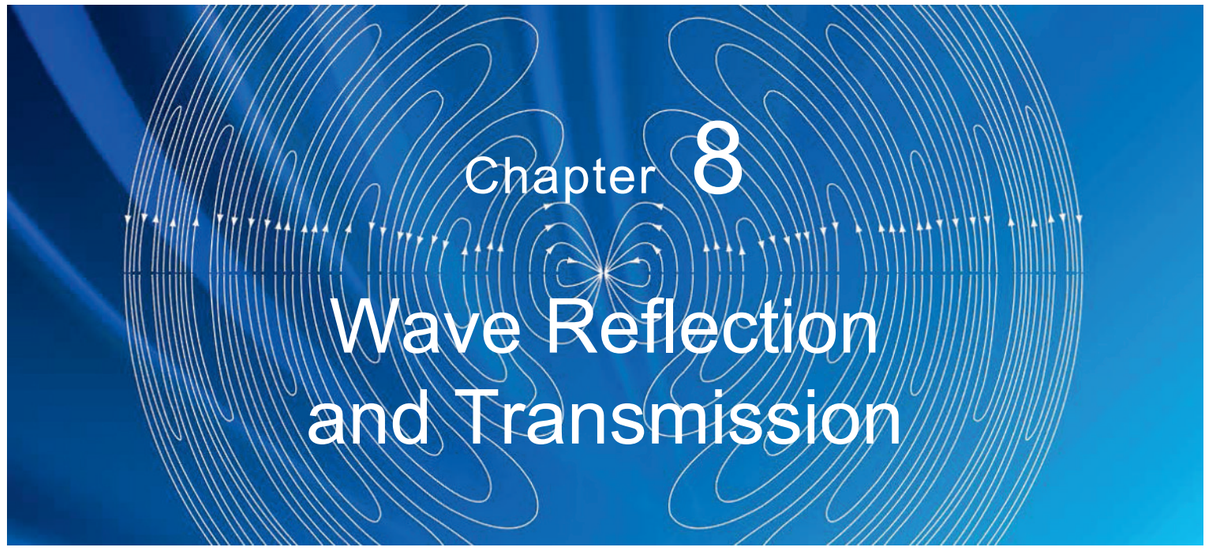
$$(w_e)_{av} = \frac{1}{T}\int_0^T w_e\, dt = \frac{1}{2T}\int_0^T \varepsilon E^2\, dt.$$

(b) 计算时均磁能密度

$$(w_m)_{av} = \frac{1}{T}\int_0^T w_m\, dt = \frac{1}{2T}\int_0^T \mu H^2\, dt.$$

(c) 显示 $(w_e)_{av} = (w_m)_{av}$。

7.42 一个科学家小组正在设计一种雷达作为探测器，用于测量南极陆地上冰层的深度。为了测量由于冰岩边界反射而产生的可检测回波，冰盖的厚度不应超过三倍趋肤深度。如果冰的情况为 $\varepsilon_r' = 3$ 和 $\varepsilon_r'' = 10^{-2}$，并且如果勘探区域的最大预期冰厚度为 1.2 公里，则雷达可使用什么频率范围？



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Objectives

Upon learning the material presented in this chapter, you should be able to:

1. Characterize the reflection and transmission behavior of plane waves incident upon plane boundaries for both normal and oblique incidence.
2. Calculate the transmission properties of optical fibers.
3. Characterize wave propagation in a rectangular waveguide.
4. Determine the behavior of resonant modes inside a rectangular cavity.

第8章

波的反射与透射

章节内容

边界处的电磁波，338 8-1 法向入射时的波反射和传输，338 TB15 激光器，347 8-2 斯涅尔定律，349 8-3 光纤，351 8-4 倾斜入射时的波反射和传输，353 TB16 条码阅读器，358 8-5 反射率和反射率透射率、361 8-6 波导、364 8-7 E 和 H 的一般关系、366 8-8 矩形波导中的 TM 模式、367 8-9 矩形波导中的 TE 模式、370 8-10 传播速度、371 8-11 空腔谐振器、374 第 8 章总结、377 问题、第378章

目标

学习本章介绍的材料后，您应该能够：

1. 表征垂直和倾斜入射的平面波在平面边界上的反射和传输行为。 2. 计算光纤的传输特性。 3. 表征波在矩形波导中的传播。 4. 确定矩形腔内谐振模式的行为。

EM Waves at Boundaries

Figure 8-1 depicts the propagation path traveled by a signal transmitted by a shipboard antenna and received by an antenna on a submerged submarine. Starting from the transmitter (denoted Tx in **Fig. 8-1**), the signal travels along a transmission line to the transmitting antenna. The relationship between the transmitter (generator) output power, P_t, and the power supplied to the antenna is governed by the transmission-line equations of Chapter 2. If the transmission line is approximately lossless and properly matched to the transmitting antenna, then all of P_t is delivered to the antenna. If the antenna itself is lossless too, it will convert all of the power P_t in the guided wave provided by the transmission line into a spherical wave radiated outward into space. The radiation process is the subject of Chapter 9. From point 1, denoting the location of the shipboard antenna, to point 2, denoting the point of incidence of the wave onto the water's surface, the signal's behavior is governed by the equations characterizing wave propagation in lossless media, which was covered in Chapter 7. As the wave impinges upon the air–water boundary, part of it is reflected by the surface while another part gets transmitted across the boundary into the water. The transmitted wave is refracted, wherein its propagation direction moves closer toward the vertical, compared with that of the direction of the incident wave. Reflection and transmission processes are treated in this chapter. Wave travel from point 3, representing a point just below the water surface, to point 4, which denotes the location of the submarine antenna, is subject to the laws of wave propagation in lossy media, which also was treated in Chapter 7. Finally, some of the power carried by the wave traveling in water towards the submarine is intercepted by the receiving antenna. The received power, P_r, is then delivered to the receiver via a transmission line. The receiving properties of antennas are covered in Chapter 9. In summary, each wave-related aspect of the transmission process depicted in **Fig. 8-1** is treated in this book, starting with the transmitter and ending with the receiver.

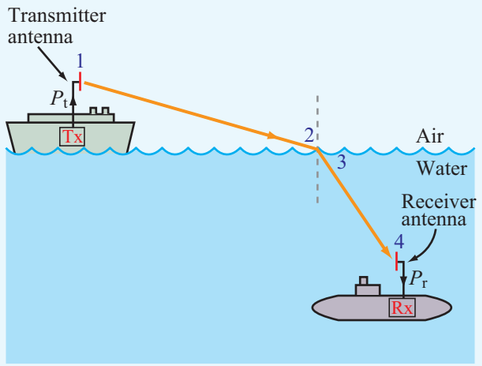


Figure 8-1 Signal path between a shipboard transmitter (Tx) and a submarine receiver (Rx).

This chapter begins by examining the reflection and transmission properties of plane waves incident upon planar boundaries and concludes with sections on waveguides and cavity resonators. Applications discussed along the way include fiber and laser optics.

8-1 Wave Reflection and Transmission at Normal Incidence

We know from Chapter 2 that, when a guided wave encounters a junction between two transmission lines with different characteristic impedances, the incident wave is partly reflected back toward the source and partly transmitted across the junction onto the other line. The same happens to a uniform plane wave when it encounters a boundary between two material half-spaces with different characteristic impedances. In fact, the situation depicted in **Fig. 8-2(b)** has an exact analogue in the transmission-line configuration of **Fig. 8-2(a)**. The boundary conditions governing the relationships between the electric and magnetic fields in **Fig. 8-2(b)** map one-to-one onto those we developed in Chapter 2 for the voltages and currents on the transmission line.

For convenience, we divide our treatment of wave reflection by and transmission through planar boundaries into two parts. In this section, we confine our discussion to the normal-incidence case depicted in **Fig. 8-3(a)**, and in Sections 8-2 to 8-4, we examine the more general oblique-incidence case depicted in **Fig. 8-3(b)**. We will show the basis for the analogy between the transmission-line and plane-wave configurations so that we may use transmission-line equivalent models, tools (e.g., Smith chart), and techniques (e.g., quarter-wavelength matching) to expeditiously solve plane-wave problems.

Before proceeding, however, we should explain the notion of rays and wavefronts and the relationship between them, as both are used throughout this chapter to represent electromagnetic waves. A ***ray*** is a line representing the direction of flow of electromagnetic energy carried by a wave; therefore, it is parallel to the propagation unit vector $\hat{\mathbf{k}}$. A ***wavefront*** is a surface across which the phase of a wave is constant; it is perpendicular to the wavevector $\hat{\mathbf{k}}$. Hence, rays are perpendicular to wavefronts. The ray representation of wave incidence, reflection, and transmission shown in **Fig. 8-3(b)** is equivalent

边界处的电磁波

图8-1描述了由舰载天线发射并由水下潜艇上的天线接收的信号所经过的传播路径。从发射机（图 8-1 中表示为 Tx）开始，信号沿着传输线传输到发射天线。发射器（发生器）输出功率 P_t 与提供给天线的功率之间的关系由第 2 章的传输线方程控制。如果传输线近似无损并且与发射天线正确匹配，则所有 P_t 都会传递到天线。如果天线本身也是无损的，它会将传输线提供的导波中的所有功率P_t转换为向外辐射到空间的球面波。辐射过程是第 9 章的主题。从点 1（表示船载天线的位置）到点 2（表示波在水面上的入射点），信号的行为受表征无损介质中的波传播的方程控制，这在第 7 章中进行了介绍。当波撞击空气-水边界时，一部分被表面反射，而另一部分则穿过边界传输到水中。发射波被折射，其中与入射波的方向相比，其传播方向更接近垂直方向。本章讨论反射和传输过程。波从代表水面下方的点 3 到代表潜艇天线位置的点 4 的传播受到有损介质中波传播定律的影响，该定律也已在第 7 章 最后，在水中传播到潜艇的波所携带的一些功率被接收天线拦截。然后，接收到的功率 P_r 通过传输线传送到接收器。第 9 章介绍了天线的接收特性。总之，本书讨论了图 8-1 中描述的传输过程中与波相关的每个方面，从发射机开始，到接收机结束。

本章首先研究入射到平面边界上的平面波的反射和传输特性，最后介绍波导和空腔谐振器。一路上讨论的应用包括光纤和激光光学。

8-1 法向入射时的波反射和透射

从第二章我们知道，当导波遇到具有不同特性阻抗的两条传输线之间的连接点时，入射波部分被反射回源，部分穿过连接点传输到另一条线路上。当均匀平面波遇到具有不同特性阻抗的两个材料半空间之间的边界时，也会发生同样的情况。事实上，图 8-2(b) 中描述的情况与图 8-2(a) 的传输线配置完全相同。图 8-2(b) 中控制电场和磁场之间关系的边界条件——映射到我们在第 2 章中针对传输线上的电压和电流开发的边界条件。

为了方便起见，我们将通过平面边界对波反射和传输的处理分为两部分。在本节中，我们将讨论限制在图 8-3(a) 所示的正入射情况下，在第 8-2 节到 8-4 节中，我们将研究图 8-3(b) 中所示的更一般的斜入射情况。我们将展示传输线和平面波配置之间的类比基础，以便我们可以使用传输线等效模型、工具（例如史密斯圆图）和技术（例如四分之一波长匹配）来快速解决平面波问题。

然而，在继续之前，我们应该解释射线和波前的概念以及它们之间的关系，因为本章通篇都使用两者来表示电磁波。射线是代表波所携带的电磁能流动方向的线；因此，它平行于传播单位向量$\hat{k}$。波前是波的相位恒定的表面；它垂直于波矢$\hat{k}$。因此，光线垂直于波前。图8-3(b)所示的波入射、反射和透射的射线表示是等效的

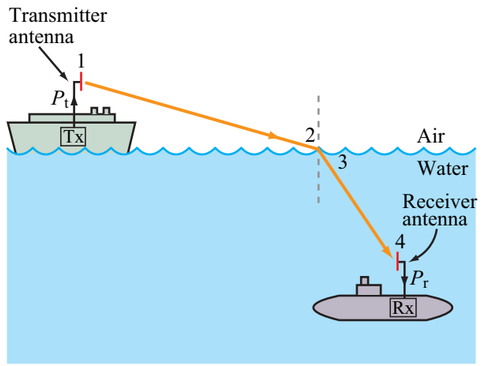
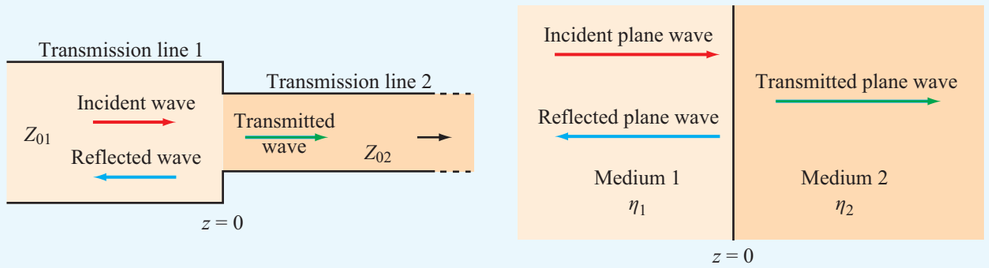
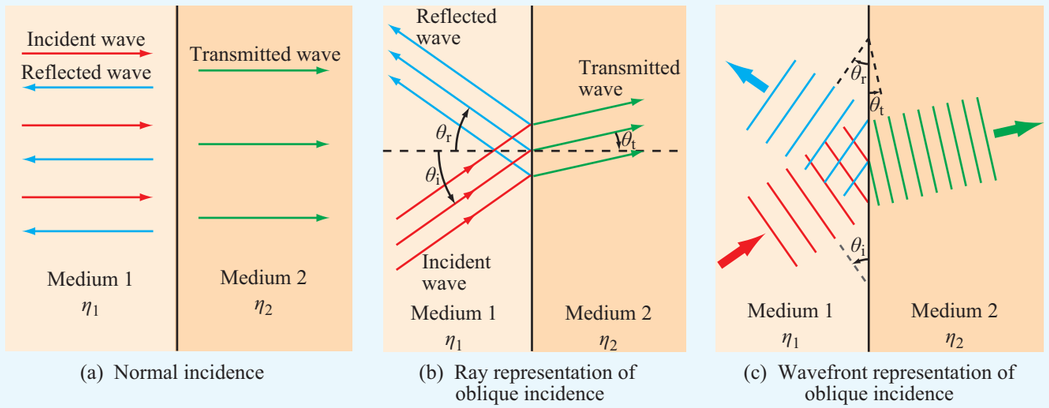


图 8-1 舰载发射器 (Tx) 和海底接收器 (Rx) 之间的信号路径。



(a) Boundary between transmission lines (b) Boundary between different media

Figure 8-2 Discontinuity between two different transmission lines is analogous to that between two dissimilar media.



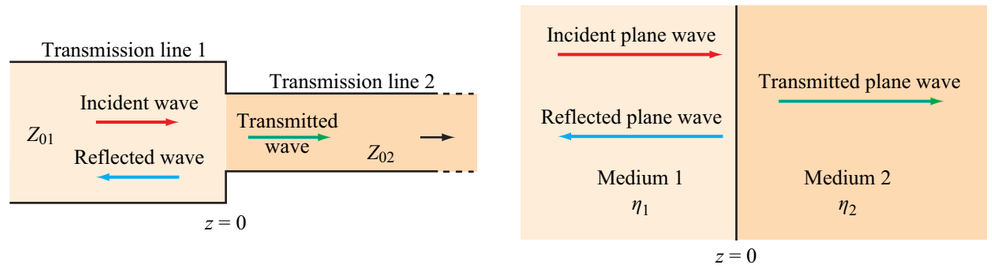
(a) Normal incidence (b) Ray representation of oblique incidence (c) Wavefront representation of oblique incidence

Figure 8-3 Ray representation of wave reflection and transmission at (a) normal incidence and (b) oblique incidence. (c) Wavefront representation of oblique incidence.

to the wavefront representation depicted in **Fig. 8-3(c)**. The two representations are complementary; the ray representation is easier to use in graphical illustrations, whereas the wavefront representation provides greater physical insight into what happens to a wave when it encounters a discontinuous boundary.

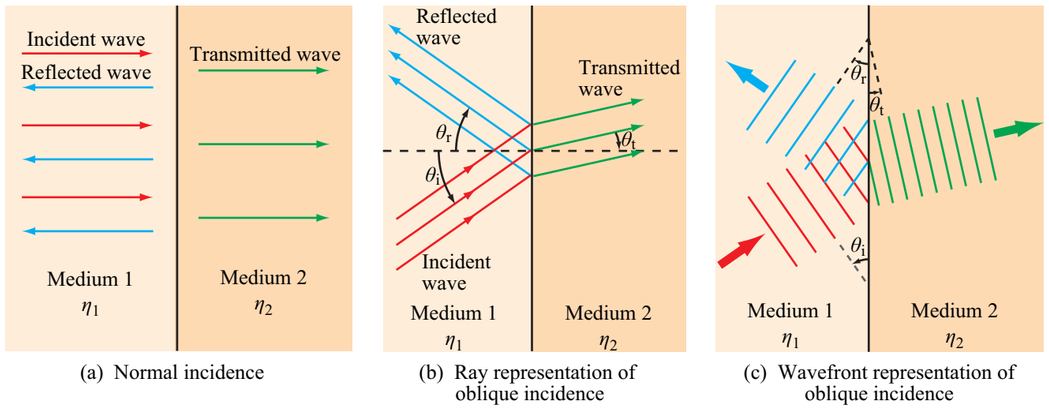
8-1.1 Boundary between Lossless Media

A planar boundary located at $z = 0$ (**Fig. 8-4(a)**) separates two lossless, homogeneous, dielectric media. Medium 1 has permittivity ε_1 and permeability μ_1 and fills the half-space $z \le 0$. Medium 2 has permittivity ε_2 and permeability μ_2 and fills the half-space $z \ge 0$. An x-polarized plane wave, with electric and magnetic fields $(\mathbf{E}^i, \mathbf{H}^i)$ propagates in medium 1 along direction $\hat{\mathbf{k}}_i = \hat{\mathbf{z}}$ toward medium 2. Reflection and transmission at the boundary at $z = 0$ result in a reflected wave, with electric and magnetic fields $(\mathbf{E}^r, \mathbf{H}^r)$, traveling along direction $\hat{\mathbf{k}}_r = -\hat{\mathbf{z}}$ in medium 1, and a transmitted wave, with electric and magnetic fields $(\mathbf{E}^t, \mathbf{H}^t)$, traveling along direction $\hat{\mathbf{k}}_t = \hat{\mathbf{z}}$ in medium 2. On the basis of the formulations developed in Sections 7-2 and 7-3 for plane waves, the three waves are described in phasor form by:



(a) Boundary between transmission lines (b) Boundary between different media

图 8-2 两条不同传输线之间的不连续性类似于两种不同介质之间的不连续性。



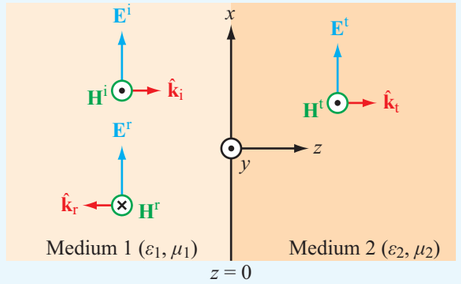
(a) Normal incidence (b) Ray representation of oblique incidence (c) Wavefront representation of oblique incidence

非图 8-3 (a) 垂直入射和 (b) 倾斜入射时波反射和透射的射线表示。 (c) 波前代表斜入射的怨恨。

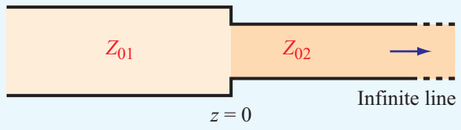
到图8-3(c)中描绘的波前表示。这两种表示是互补的；射线表示更容易在图形插图中使用，而波前表示可以更深入地了解波遇到不连续边界时会发生什么。

8-1.1 无损媒体之间的界限

位于 z = 0 处的平面边界（图 8-4(a)）将两种无损、均匀的介电介质分开。介质 1 具有介电常数 ε_1 和磁导率 μ_1 并填充半空间 $z \le 0$。介质 2 具有介电常数 ε_2 和磁导率 μ_2，并填充半空间 $z \ge 0$。具有电场和磁场 (E^i、H^i) 的 x 偏振平面波在介质 1 中沿方向 $\hat{k}_i = \hat{z}$ 向介质 2 传播。在 z = 0 边界处的反射和透射产生具有电场和磁场 (E^r, H^r) 的反射波，在介质 1 中沿 $k_r = -\hat{z}$ 方向传播，以及具有电场和磁场 (E^t, H^t) 的透射波，在介质 2 中沿 $\hat{k}_t = \hat{z}$ 方向传播。 7-2和7-3对于平面波，这三个波以相量形式描述为：



(a) Boundary between dielectric media



(b) Transmission-line analogue

Figure 8-4 The two dielectric media separated by the *x*–*y* plane in (a) can be represented by the transmission-line analogue in (b).

Incident Wave

$$\widetilde{\mathbf{E}}^{\mathrm{i}}(z) = \hat{\mathbf{x}} E_0^{\mathrm{i}} e^{-jk_1 z}, \tag{8.1a}$$

$$\widetilde{\mathbf{H}}^{\mathrm{i}}(z) = \hat{\mathbf{z}} \times \frac{\widetilde{\mathbf{E}}^{\mathrm{i}}(z)}{\eta_1} = \hat{\mathbf{y}} \frac{E_0^{\mathrm{i}}}{\eta_1} e^{-jk_1 z}. \tag{8.1b}$$

Reflected Wave

$$\widetilde{\mathbf{E}}^{\mathrm{r}}(z) = \hat{\mathbf{x}} E_0^{\mathrm{r}} e^{jk_1 z}, \tag{8.2a}$$

$$\widetilde{\mathbf{H}}^{\mathrm{r}}(z) = (-\hat{\mathbf{z}}) \times \frac{\widetilde{\mathbf{E}}^{\mathrm{r}}(z)}{\eta_1} = -\hat{\mathbf{y}} \frac{E_0^{\mathrm{r}}}{\eta_1} e^{jk_1 z}. \tag{8.2b}$$

Transmitted Wave

$$\widetilde{\mathbf{E}}^{\mathrm{t}}(z) = \hat{\mathbf{x}} E_0^{\mathrm{t}} e^{-jk_2 z}, \tag{8.3a}$$

$$\widetilde{\mathbf{H}}^{\mathrm{t}}(z) = \hat{\mathbf{z}} \times \frac{\widetilde{\mathbf{E}}^{\mathrm{t}}(z)}{\eta_2} = \hat{\mathbf{y}} \frac{E_0^{\mathrm{t}}}{\eta_2} e^{-jk_2 z}. \tag{8.3b}$$

The quantities E_0^{i}, E_0^{r}, and E_0^{t} are, respectively, the amplitudes of the incident, reflected, and transmitted electric fields at $z = 0$ (the boundary between the two media). The wavenumber and intrinsic impedance of medium 1 are $k_1 = \omega\sqrt{\mu_1 \varepsilon_1}$ and $\eta_1 = \sqrt{\mu_1/\varepsilon_1}$, and those for medium 2 are $k_2 = \omega\sqrt{\mu_2 \varepsilon_2}$ and $\eta_2 = \sqrt{\mu_2/\varepsilon_2}$.

The amplitude E_0^{i} is imposed by the source responsible for generating the incident wave; therefore, it is assumed known. Our goal is to relate E_0^{r} and E_0^{t} to E_0^{i}. We do so by applying boundary conditions for the total electric and magnetic fields at $z = 0$. According to **Table 6-2**, the tangential component of the total electric field is always continuous across a boundary between two contiguous media, and in the absence of current sources at the boundary, the same is true for the total magnetic field. In the present case, the electric and magnetic fields of the incident, reflected, and transmitted waves are all tangential to the boundary.

The total electric field $\widetilde{\mathbf{E}}_1(z)$ in medium 1 is the sum of the electric fields of the incident and reflected waves, and a similar statement applies to the magnetic field $\widetilde{\mathbf{H}}_1(z)$. Hence,

Medium 1

$$\widetilde{\mathbf{E}}_1(z) = \widetilde{\mathbf{E}}^{\mathrm{i}}(z) + \widetilde{\mathbf{E}}^{\mathrm{r}}(z) = \hat{\mathbf{x}}(E_0^{\mathrm{i}} e^{-jk_1 z} + E_0^{\mathrm{r}} e^{jk_1 z}), \tag{8.4a}$$

$$\widetilde{\mathbf{H}}_1(z) = \widetilde{\mathbf{H}}^{\mathrm{i}}(z) + \widetilde{\mathbf{H}}^{\mathrm{r}}(z) = \hat{\mathbf{y}} \frac{1}{\eta_1} (E_0^{\mathrm{i}} e^{-jk_1 z} - E_0^{\mathrm{r}} e^{jk_1 z}). \tag{8.4b}$$

With only the transmitted wave present in medium 2, the total fields are

Medium 2

$$\widetilde{\mathbf{E}}_2(z) = \widetilde{\mathbf{E}}^{\mathrm{t}}(z) = \hat{\mathbf{x}} E_0^{\mathrm{t}} e^{-jk_2 z}, \tag{8.5a}$$

$$\widetilde{\mathbf{H}}_2(z) = \widetilde{\mathbf{H}}^{\mathrm{t}}(z) = \hat{\mathbf{y}} \frac{E_0^{\mathrm{t}}}{\eta_2} e^{-jk_2 z}. \tag{8.5b}$$

At the boundary ($z = 0$), the tangential components of the electric and magnetic fields are continuous. Hence,

$$\widetilde{\mathbf{E}}_1(0) = \widetilde{\mathbf{E}}_2(0) \quad \text{or} \quad E_0^{\mathrm{i}} + E_0^{\mathrm{r}} = E_0^{\mathrm{t}}, \tag{8.6a}$$

$$\widetilde{\mathbf{H}}_1(0) = \widetilde{\mathbf{H}}_2(0) \quad \text{or} \quad \frac{E_0^{\mathrm{i}}}{\eta_1} - \frac{E_0^{\mathrm{r}}}{\eta_1} = \frac{E_0^{\mathrm{t}}}{\eta_2}. \tag{8.6b}$$

Solving these equations for E_0^{r} and E_0^{t} in terms of E_0^{i} gives

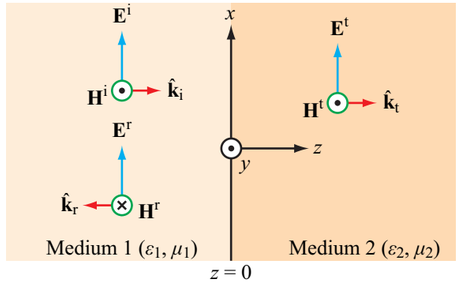
$$E_0^{\mathrm{r}} = \left(\frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}\right) E_0^{\mathrm{i}} = \Gamma E_0^{\mathrm{i}}, \tag{8.7a}$$

$$E_0^{\mathrm{t}} = \left(\frac{2\eta_2}{\eta_2 + \eta_1}\right) E_0^{\mathrm{i}} = \tau E_0^{\mathrm{i}}, \tag{8.7b}$$

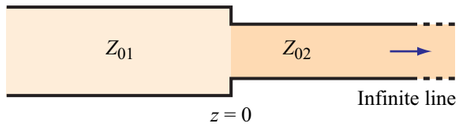
where

$$\Gamma = \frac{E_0^{\mathrm{r}}}{E_0^{\mathrm{i}}} = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} \quad \textbf{(normal incidence)}, \tag{8.8a}$$

$$\tau = \frac{E_0^{\mathrm{t}}}{E_0^{\mathrm{i}}} = \frac{2\eta_2}{\eta_2 + \eta_1} \quad \textbf{(normal incidence)}. \tag{8.8b}$$



(a) Boundary between dielectric media



(b) Transmission-line analogue

图 8-4 (a) 中由 x-y 平面分隔开的两种介电介质可以用 (b) 中的传输线模拟来表示。

事件波

$$\widetilde{\mathbf{E}}^{\mathrm{i}}(z) = \hat{\mathbf{x}} E_0^{\mathrm{i}} e^{-jk_1 z}, \tag{8.1a}$$

$$\widetilde{\mathbf{H}}^{\mathrm{i}}(z) = \hat{\mathbf{z}} \times \frac{\widetilde{\mathbf{E}}^{\mathrm{i}}(z)}{\eta_1} = \hat{\mathbf{y}} \frac{E_0^{\mathrm{i}}}{\eta_1} e^{-jk_1 z}. \tag{8.1b}$$

反射波

$$\widetilde{\mathbf{E}}^{\mathrm{r}}(z) = \hat{\mathbf{x}} E_0^{\mathrm{r}} e^{jk_1 z}, \tag{8.2a}$$

$$\widetilde{\mathbf{H}}^{\mathrm{r}}(z) = (-\hat{\mathbf{z}}) \times \frac{\widetilde{\mathbf{E}}^{\mathrm{r}}(z)}{\eta_1} = -\hat{\mathbf{y}} \frac{E_0^{\mathrm{r}}}{\eta_1} e^{jk_1 z}. \tag{8.2b}$$

传输波

$$\widetilde{\mathbf{E}}^{\mathrm{t}}(z) = \hat{\mathbf{x}} E_0^{\mathrm{t}} e^{-jk_2 z}, \tag{8.3a}$$

$$\widetilde{\mathbf{H}}^{\mathrm{t}}(z) = \hat{\mathbf{z}} \times \frac{\widetilde{\mathbf{E}}^{\mathrm{t}}(z)}{\eta_2} = \hat{\mathbf{y}} \frac{E_0^{\mathrm{t}}}{\eta_2} e^{-jk_2 z}. \tag{8.3b}$$

量 E_0^{i}、E_0^{r} 和 E_0^{t} 分别是 $z = 0$（两种介质之间的边界）处的入射电场、反射电场和透射电场的振幅。介质 1 的波数和本征阻抗为 $k_1 = \omega\sqrt{\mu_1 \varepsilon_1}$ 且 $\eta_1 = \sqrt{\mu_1/\varepsilon_1}$，介质 2 的为 $k_2 = \omega\sqrt{\mu_2 \varepsilon_2}$ 和 $\eta_2 = \sqrt{\mu_2/\varepsilon_2}$。振幅 E_0^{i} 由负责生成入射波的源施加；因此，假设已知。我们的目标是将 E_0^{r} 和 E_0^{t} 与 E_0^{i} 联系起来。为此，我们对 $z = 0$ 处的总电场和磁场应用边界条件。根据表 6-2，总电场的切向分量在两个相邻介质之间的边界上始终是连续的，并且在边界处没有电流源的情况下，总磁场也是如此。在本例中，入射波、反射波和透射波的电场和磁场均与边界相切。

介质 1 中的总电场 $\widetilde{\mathrm{E}}_1(z)$ 是入射波和反射波的电场之和，类似的说法也适用于磁场 $\widetilde{\mathrm{H}}_1(z)$。因此，

中1

$$\widetilde{\mathbf{E}}_1(z) = \widetilde{\mathbf{E}}^{\mathrm{i}}(z) + \widetilde{\mathbf{E}}^{\mathrm{r}}(z) = \hat{\mathbf{x}}(E_0^{\mathrm{i}} e^{-jk_1 z} + E_0^{\mathrm{r}} e^{jk_1 z}), \tag{8.4a}$$

$$\widetilde{\mathbf{H}}_1(z) = \widetilde{\mathbf{H}}^{\mathrm{i}}(z) + \widetilde{\mathbf{H}}^{\mathrm{r}}(z) = \hat{\mathbf{y}} \frac{1}{\eta_1} (E_0^{\mathrm{i}} e^{-jk_1 z} - E_0^{\mathrm{r}} e^{jk_1 z}) \text{。} \tag{8.4b}$$

当介质 2 中仅存在发射波时，总场为

中2

$$\widetilde{\mathbf{E}}_2(z) = \widetilde{\mathbf{E}}^{\mathrm{t}}(z) = \hat{\mathbf{x}} E_0^{\mathrm{t}} e^{-jk_2 z}, \tag{8.5a}$$

$$\widetilde{\mathbf{H}}_2(z) = \widetilde{\mathbf{H}}^{\mathrm{t}}(z) = \hat{\mathbf{y}} \frac{E_0^{\mathrm{t}}}{\eta_2} e^{-jk_2 z}. \tag{8.5b}$$

在边界 ($z = 0$) 处，电场和磁场的切向分量是连续的。因此，

$$\widetilde{\mathbf{E}}_1(0) = \widetilde{\mathbf{E}}_2(0) \quad \text{or} \quad E_0^{\mathrm{i}} + E_0^{\mathrm{r}} = E_0^{\mathrm{t}}, \tag{8.6a}$$

$$\widetilde{\mathbf{H}}_1(0) = \widetilde{\mathbf{H}}_2(0) \quad \text{or} \quad \frac{E_0^{\mathrm{i}}}{\eta_1} - \frac{E_0^{\mathrm{r}}}{\eta_1} = \frac{E_0^{\mathrm{t}}}{\eta_2}. \tag{8.6b}$$

根据 E_0^{i} 求解这些方程，得到 E_0^{r} 和 E_0^{t}

$$E_0^{\mathrm{r}} = \left(\frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}\right) E_0^{\mathrm{i}} = \Gamma E_0^{\mathrm{i}}, \tag{8.7a}$$

$$E_0^{\mathrm{t}} = \left(\frac{2\eta_2}{\eta_2 + \eta_1}\right) E_0^{\mathrm{i}} = \tau E_0^{\mathrm{i}}, \tag{8.7b}$$

在哪里

$$\Gamma = \frac{E_0^{\mathrm{r}}}{E_0^{\mathrm{i}}} = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} \quad \textbf{(normal incidence)}, \tag{8.8a}$$

$$\tau = \frac{E_0^{\mathrm{t}}}{E_0^{\mathrm{i}}} = \frac{2\eta_2}{\eta_2 + \eta_1} \quad \textbf{(normal incidence)}. \tag{8.8b}$$